

UNIT-IV

Search Statements&Binding:

The Search statements are the statements of an Information and generated by users to specify the concepts, they are trying to locate in items

- It uses traditional Boolean logic &or Natural Language
- While generating the search statement, the user have the ability to weight or assign an Important to different concepts in the statements
- At this point the binding is to the vocabulary& past experiences of the user
- The search statement is the users attempt to specify the conditions needed to subset logically the total Item Space to that cluster of Items that contains the information needed by the user
- The next level of binding comes when the search statements is parsed for use by a specific search system
- The search System translates the query to its own Meta Language
- This process is similar to the Indexing of Item processes

Ex:-

Statistical systems determine the processing tokens of Interest and the weights assigned to each processing token based up on the frequency of occurrence from the Search Statement

- Natural Language Systems determine the Syntactical and Discourse Semantics using Algorithms Similar to those used in Indexing
- Concept Systems map the search statements to the set of concepts used to Index Items
- Some of the Statistics used in weighing are based upon the current contents of the Database
- Some examples are Documents frequency &Total Frequency for a Specific term
- Parenthesis are used in the second binding step to Indicate Expansion by a Thesaurus

INPUT	Binding
"Find me information on the impact of the oil spills in Alaska on the price of oil"	User search statement using vocabulary of user
impact, oil (petroleum), spills (accidents), Alaska, price (cost, value)	Statistical system binding extracts processing tokens
impact (.308), oil (.606), petroleum (.65), spills (.12), accidents (.23), Alaska (.45), price (.16), cost (.25), value (.10)	Weights assigned to search terms based upon inverse document frequency algorithm and database

Figure 7.1 Examples of Query Binding

The length of search Statements directly affect the ability of Information Retrieval Systems to find relevant Items

The longer the search query, the easier it is for the system to find items

Similarity Measures and Ranking

The Searching is concerned with calculating the Similarity between a users Search Statement and the items in the Database

- Restricting the similarity Measure to Passages gains Significant Precision with Impact on recall
- Once Items are Identified as possibly relevant to the users query, It is best to present the most likely relevant items first Ranking is a Scalar number that represents how similar an Item is to the Query
- Searching is Concerned with calculating the Similarity between a users search statement and the items in the Database
- Variety of different similarity measures can be used to calculate the similarity the item and the search Statement

- A Characteristic of a similarity formula is that the results of the Formula Increase

Increases as the Items become More Similar

- The value is Zero if the Items are totally Dissimilar
- $SIM(Item\ i, Item\ j) = z(Term_{ix})(Term_{ja})$
- This formula uses the summation of the product of the various terms of two Items when treating the Index as a vector
- The problem with the simple Measures in the normalization needed to account for variances in the length of Items

Similarity Formula by Salton in SMART System

To determine the weight an Item has with respect to the Search Statement, The Cosine Formula is used to calculate the distance between the vector for the Item and the vector for the Query

$$SIM(DOC_i, QUERY_j) = \frac{\sum_{k=1}^n (DOC_{i,k} * QTERM_{j,k})}{\sqrt{\sum_{k=1}^n (DOC_{i,k})^2 * \sum_{k=1}^n (QTERM_{j,k})^2}}$$

The DICE Formula

The Measure Simplifies the denominator from the Jaccard Formula and Introduces a factor of 2 in the Numerator

- The Normalization in the Dice Formula is also Invariant to the number of terms in common

$$SIM(DOC_i, QUERY_j) = \frac{2 * \sum_{k=1}^n (DOC_{i,k} * QTERM_{j,k})}{\sum_{k=1}^n DOC_{i,k} + \sum_{k=1}^n QTERM_{j,k}}$$

The use of a similarity Algorithm returns the complete Database as Search Results

Many of items have Similarity Close or equal to zero

- The Threshold defines the Items in the resultant Hit File from the Query

Query Threshold Process:

Vector:- American, geography, Lakes, Mexico, Painter, oil, Reserve, Subject

DOC1:- geography of Mexico suggests oil reserves are available

Vector(0,1,0,2,0,3,1,0)

DOC2:- American Geography has takes available everywhere

vector(1,3,0,0,0,0,0,0)

DOC3:-Painters Suggest Mexico Lakes as Subjects Vector(0,0,1,3,3,0,0,2)

Based on Existing of Occurrence &Based frequency Occurrence numerical values

Relevance Feedback:

The Thesaurus and Semantic networks provide Utility in generally

Expanding a users search statement to Include Potential related Search terms

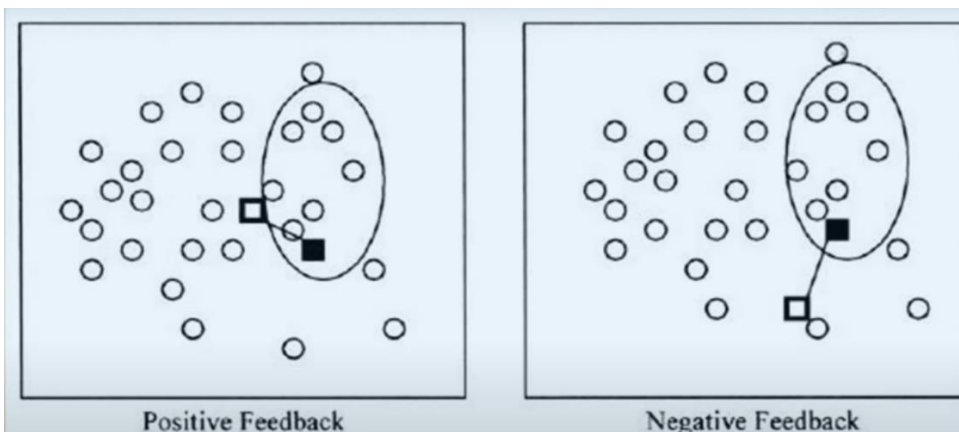
- But this still does not Correlate to the vocabulary used by the authors that contributes to a particular Database
- There is also a significant risk that the Thesaurus does not Include the latest words being used
- In an Interactive System, users can Manually Modify an Inefficient Query as well as the system Automatically Expand the Query Via Thesaurus
- The Relevant Items are used to reweight the Existing Query terms and possible Expand the users Search Statement with terms
- The relevance feedback concept was that new Query Should be based on the old Query Modified to Increase the weight of terms in relevant Items and decrease the weight of terms that are in non-relevant Items

The formula used for Relevance Feedback

$$Q_n = Q_o + \frac{1}{r} \sum_{i=1}^r DR_i - \frac{1}{nr} \sum_{j=1}^{nr} DNR_j$$

where

Q_n = the revised vector for the new query
 Q_o = the original query
 r = number of relevant items
 DR_i = the vectors for the relevant items
 nr = number of non-relevant items
 DNR_j = the vectors for the non-relevant items.



Impact of Relevance Feedback

Selective Dissemination of Information Search:

The Selective Dissemination of Information, frequently Called dissemination Systems are becoming more prevalent with the growth of Internet

- A Dissemination system is sometimes called as Push Pull system
- Push is nothing but whenever you search statement ,Pull is nothing but Extract on Information based on Statement(or)back to the Information
- In a dissemination system, the user defines a profile and as new Information is added to the system, it is Automatically compared to the user's Profile
- If it is considered a Match, it is asynchronously sent to the user mail file
- The difference between the two functions lie in the Dynamic nature of the profiling process, The size and diversity of the search statements and the number of Simultaneous searches per Item
- In the Search System an Existing Database Exists
- These can be used for weighting factors in the Indexing process and the similarity Comparison
- Profiles are relatively static search statements that cover a diversity of topics
- One of the first commercial search Techniques for dissemination was the logic on Message Dissemination System(LMDS)
- The system originated from a system created by chase, Rosen and Wallace(CRW)
- It was designed for speed to support the search of Thousands of profiles with the items arriving every 20 seconds
- Another approach to dissemination uses a Statistical Classification Technique and Explicit Error Minimization to determine the decision criteria for selecting items for a particular profile

Weighted Searches of Boolean Systems:

The two major approaches to generating Queries are Boolean and Natural Language Queries are easily represented within statistical Models and are usable by the Similarity Measures Discussed

- The Issues arise when Boolean Queries are Associated with weighted Index Systems
- Some of the Issues are Associated with how the logic (AND,OR, NOT) Operators Function with Weighted values and How weights are Associated with the Query terms
- If the operators are Interpreted in their normal Interpretation, they act too restrictive or too general AND and OR Operators Respectively
- Closely related to the strict definition problem is the lack of ranking that is missing from a pure Boolean Process
- Some of the early work addressing this problem recognized the Fuzziness associated with mixing Boolean and weighted systems(Brookstein-78,Brookstein-80)
- To Integrate the Boolean and weighted systems Model (Fuzzy set=Fuzzy sets are sets whose Elements have degrees of membership
- Fuzzy sets were Introduced “Zadeh” Fuzzylogic was Introduced with 1965 Mathematician
- The degree of Membership for AND and OR operations are defined

$$DEG_{A \cap B} = \min(DEG_A, DEG_B)$$

$$DEG_{A \cup B} = \max(DEG_A, DEG_B)$$

$$SIM(QUERY_{OR}, DOC) = C_{OR1} * \max(DOC1_1, DOC2, \dots, DOC_n) + C_{OR2} * \min(DOC1, DOC2, \dots, DOC_n)$$

$$SIM(QUERY_{AND}, DOC) = C_{AND1} * \min(DOC1, DOC2, \dots, DOC_n) + C_{AND2} * \max(DOC1_1, DOC2, \dots, DOC_n)$$

Considering all item weighs versus the Maximum/Minimum approach the Similarity Measure is Calculated as

$$SIM(QUERY DOC) = \sum_{i=1}^n r^{i-1} d_i / \sum_{i=1}^n r^{i-1}$$

$$Q_{OR} = (A_1, a_1) \text{ OR } (A_2, a_2) \text{ OR } \dots \text{ OR } (A_n, a_n)$$

$$Q_{AND} = (A_1, a_1) \text{ AND } (A_2, a_2) \text{ AND } \dots \text{ AND } (A_n, a_n)$$

Searching the Internet & Hypertext

The Internet has multiple different mechanisms that are the basis for searching of items. The primary techniques are associated with servers on the Internet that create indexes of items on the Internet and allow search of them. Some of the most commonly used nodes are YAHOO, AltaVista and Lycos. In all of these systems there are active processes that visit a large number of Internet sites and retrieve textual data which they index.

The primary design decisions are on the level to which they retrieve data and their general philosophy on user access. LYCOS (<http://www.lycos.com>) and AltaVista automatically go out to other Internet sites and return the text at the sites for automatic indexing (<http://www.altavista.digital.com>). Lycos returns home pages from each site for automatic indexing while Altavista indexes all of the text at a site.

- The retrieved text is then used to create an index to the source items storing the Uniform Resource Locator (URL) to provide to the user to retrieve an item. All of the systems use some form of ranking algorithm to assist in display of the retrieved items. The algorithm is kept relatively simple using statistical information on the occurrence of words within the retrieved text
- Closely associated with the creation of the indexes is the technique for accessing nodes on the Internet to locate text to be indexed. This search process is also directly available to users via Intelligent Agents. Intelligent Agents provide the capability for a user to specify an information, need which will be used by the Intelligent Agent as it independently moves between Internet sites locating information of interest.

There are six key characteristics of intelligent agents

Autonomy:

The search agent must be able to operate without interaction with a human agent. It must have control over its own internal states and make independent decisions. This implies a search capability to traverse information sites based upon pre-established criteria collecting potentially relevant information.

Communications Ability:

The agent must be able to communicate with the information sites as it traverses them. This implies a universally accepted language defining the external interfaces

Capacity for Cooperation:

This concept suggests that intelligent agents need to cooperate to perform mutually beneficial tasks.

Capacity for Reasoning:

There are three types of reasoning scenarios

Rule-based - where user has defined a set of conditions and actions to be taken

Knowledge-based - where the intelligent agents have stored previous conditions and actions taken which are used to deduce future actions

Artificial evolution based - where intelligent agents spawn new agents with higher logic capability to perform its objectives.

Adaptive Behavior - closely tied to 1 and 4 , adaptive behavior permits the intelligent agent to assess its current state and make decisions on the actions it should take

Trustworthiness- The user must trust that the intelligent agent will act on the user's behalf to locate information that the user has access to and is relevant to the user.

Introduction of Information Visualization:

The Visualization is the transformation of Information into a visual form which enables the user to observe and Understand the Information

- The functions that are available with Electronic display and Visualization of Data

- Modify representations of Data and Information or Display Conditions(Changing colors scales)
- Use the same representation while showing changes in data
- Animate the display to show changes in space &Time
- Create Hyperlinks under user control to establish relationships between data
- The Information visualization addresses how the results of search may be optimally display to the users to facilitate their understanding of what the search has provided and their selection of most likely Items of Interest to read Cognitive(The action or process of acquiring Knowledge and understanding through Experience &The Sense)
- The Engineering derives design principles for Visualization techniques
- It is Attention memory and information processing of the human visual system

There are many areas that information visualization and presentation can help the user:

- A. The reduce the amount of time to understand the results of a search and likely clusters of relevant information
 - B. yield information that comes from the relationships between items versus treating each item as independent
 - C. perform simple actions that produce sophisticated information search functions
- Visualization can be divided into two broad classes
 - Link Visualization
 - Attribute visualization

Link Visualization:

It displays relationships among items

Attribute Visualization:

It reveals Content relationships across Large Numbers of Items

Cognition:- cognition means to Store the Information(The action or process of acquiring knowledge and understanding through Thought, Experience& the Senses)

Preception:- Preception means Receiving Information(the ability to see, hear, or become aware of something through the senses)

- The user Machine Interface has primarily focused on a paradigm of a typewriter
- As computers display became Man-Machine Interfaces focused on treating the display as an Extension of paper with the focus on consistency of operations
- The Advent of WLMP(Windows, Icons, Menus and Pointers)Interfaces and the Evolution of the user what is talking place in the computer Environment
- Extending the HCL(Human Computer Interface) to Improve the Information Flow. Thus reducing wasted user overhead in Locating needed Information
- Although the Major focus is on enhanced visualization of Information,other Senses are along being looked at for Future Interfaces
- The Audio Sense has been always been part of Simple Alerts in Computers
- The sounds are now being replaced by Speech in both Input& output Interfaces
- The Touch Senses is being addressed in the Experiments using Virtual Realty
- A Significant portion of the brain is devoted to vision and supports the Maximum Information transfer Function from the environment to a humanbeing

Other Measures:

- **Proximity**-Near by figures are grouped together
- **Similarity**- Similar figures are grouped together
- **Continuity**-figures are Interpreted as Smooth Continuous patterns rather than discontinuous Concatenations of Shapes
(Eg:-Circle with its Diameter drawn is perceived as two continuous shapes, a circle& a Line, Versus two half Circles Concatenated together)

Closure-gaps within a figure are filled into create a Whole(Eg:-using Dashed lines to represent a Square does not prevent understanding it a square)

Connectedness:- Uniform and Linked Spots,Lines or Areas are perceived as a Single Unit

- Shifting the Information Processing Load from Slower Cognitive processes to faster perceptual Systems Significantly Improves the Information carrying Interfaces between Humans &computers

Information Visualization Technologies

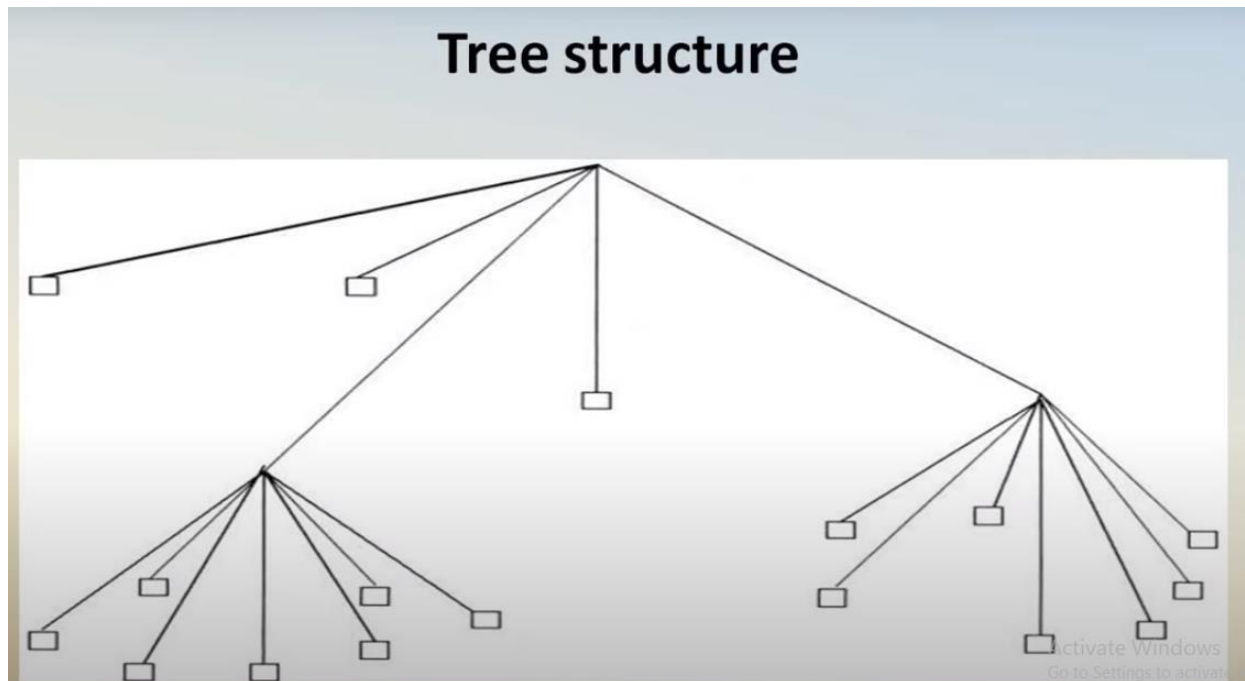
The main focus on Information Retrieval Systems are as follows

- The Investigating how to display the results of Searches Effectively, the structured Data from DBMS
- The Results of link analysis Correlating Data
- The goals of display the result in search is divided into 2 Major Classes
 - Document Clustering
 - Search Statement Analysis
- Visualization tools helps to display the Clusters in their Size &Topic to provide users to Navigate to items of Interest
- The Displaying the total set of terms
- Including additional terms from relevance feedback thesaurus Expansion Along with Documents retrieved and Indicate the Importance of the terms to the retrieval and Ranking process various information Visualization Technologies are as follows
 - Tree Structure
 - Cone-Tree
 - Perspective Wall
 - Tree Maps
 - Envision System
 - Document content Analysis &Retrieval System(DCARS)
 - City Space

Tree Structure:

A Tree Structure is useful in representing Information that ranges overtime

- The Constituents of a larger Unit(ex:- Organization Structures, Mechanical device Definitions)
- The Aggregates from the higher to lower level(Eg: Hierarchical clustering of Documents)



CONE-TREE:

The Cone-Tree is a 3-Dimensional Representation of Data

Where one node of the tree is represented at the Apex

And all Information Subordinate to it is arranged in a Circular Structure at its base

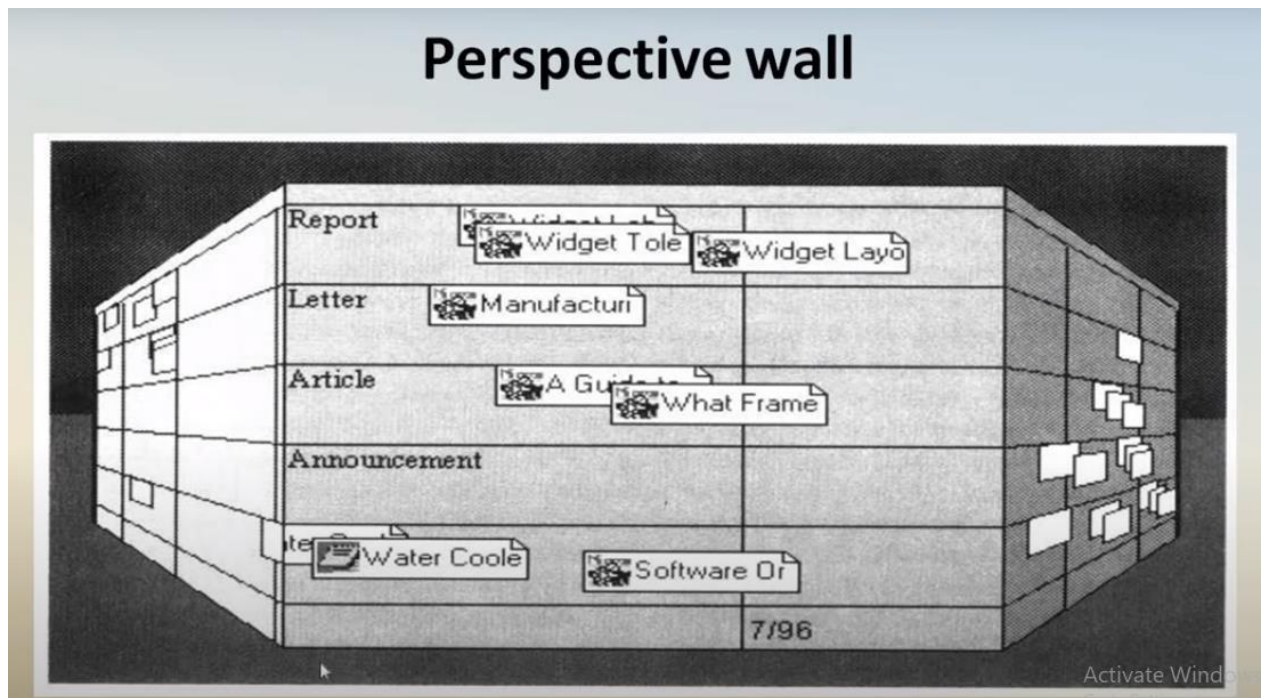
Any Child node may also be the Parent of another Cone

Selecting a particular node, rotates it to the front of the display

Perspective Wall:-

The perspective wall divides the Information into three Visual Areas

This allows the user to keep all of the Information in Perspective while Focusing on Particular Area

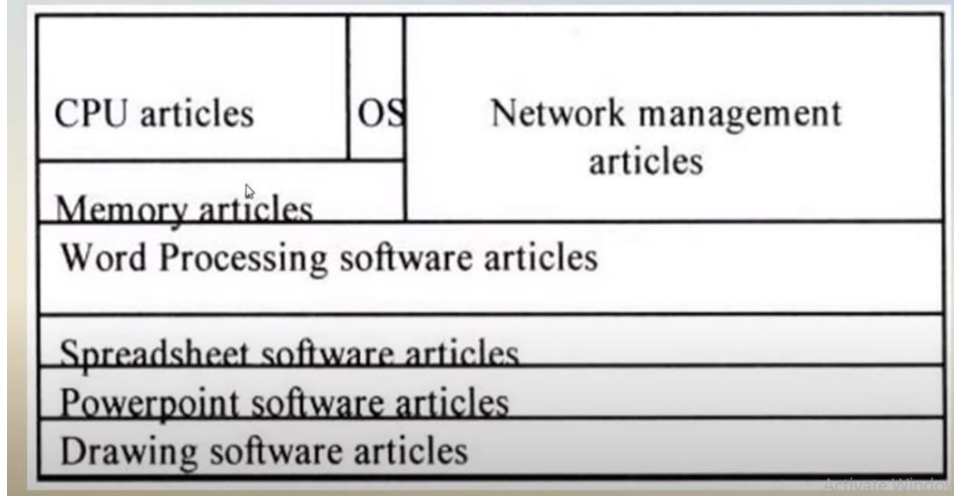


Tree Maps:-

This Technique Makes Maximum use of the display Screen Space by using Rectangular boxes that are Recursively subdivided and based on Parent-Child relationships between the Data

- The CPU
- OS
- Memory and
- The Network Management articles are all related to a general category of computer operating Systems
- The Computer Applications which are shown in the rest of the Figure

Tree maps(Johnson-91)

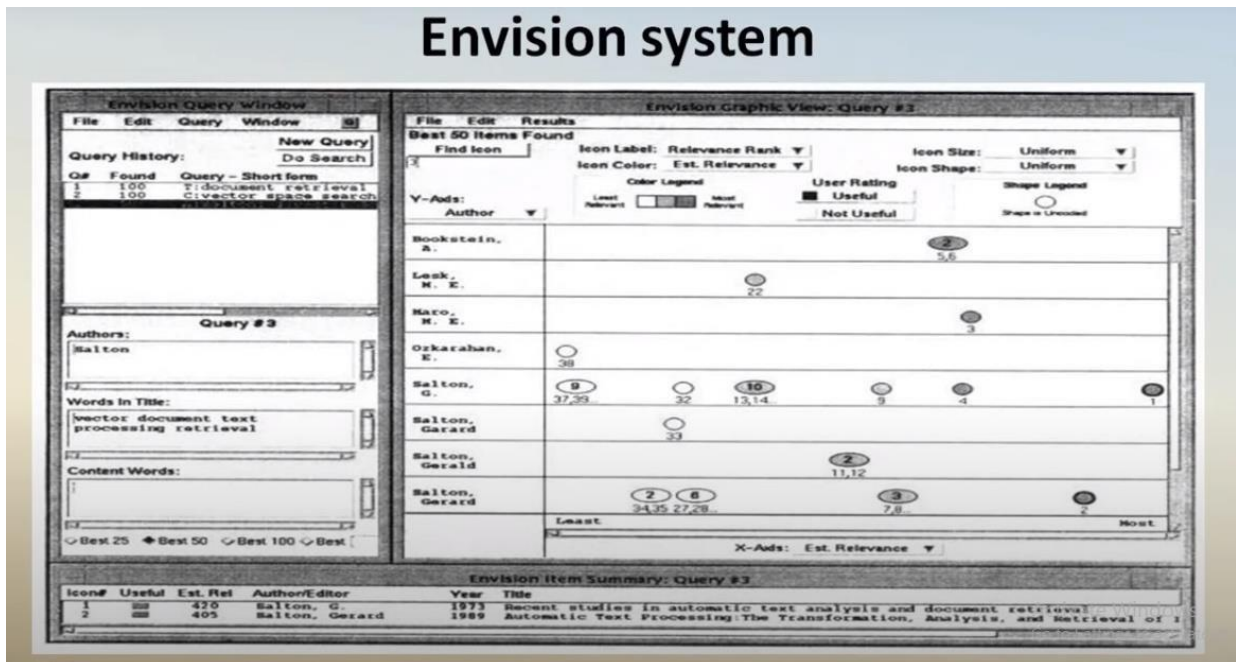


Envision System:-

The Envision System not only displays the relevance rank and estimated relevance of each Item found by a query, but also Simultaneously Presents other query Information

The design is Intentionally graphical and simple using two Dimensional Visualization

Envision system



Document content Analysis & Retrieval System (DCARS):-

- The Document Content Analysis and Retrieval System DCARS being developed by Calspan Advanced Technology Center
- Their System is designed to Augment the Retrieval ware Search Product
- They display the query results as a histogram with the items as rows and each terms Contribution to the Selection Indicated by the Width of a Title bar on the row
- DCARS Provides a friendly user Interface that Indicates why a particular Item was Found but it is much harder to use the Information in determining how to modify Search Statements to Improve them

CITY SCAPE:

- This Representation is widely used for both hierarchical and Network related Information is the Cityscape which uses the Metaphor of Movement within a city
- In lieu of using hills as in the terrain approach Sky Scrapers Represent the theme Areas

